



The Ephemeris



December 2022

Volume 33 Number 04- The Official Publication of the San Jose Astronomical Association

December 2022 - March 2023 Events

Schedule subject to change

Please check our website for up to date information.

Board & Speaker Talk (online)

Saturday 12/10, 1/7, 2/4, 3/4

Board Meetings: 6 -7:30pm

Speaker Talk: 8-9:30pm

Solar Sunday (1:30-3:30pm)

Sunday 12/11, 1/22, 2/12, 3/5

Intro to the Night Sky & In-Town Star Party

Friday 12/16, 1/13, 2/17, 3/24

RCDO Starry Nights Star Party

Saturday 12/17, 1/14, 2/11, 3/11

Imaging SIG Mtg

Tuesday 12/18, 1/17, 2/21, 3/21

Hands-on Imaging (at Little Uvas OSP)

Saturday 1/21, 2/18, 3/18

Binocular Star Gazing

Will resume in Summer 2023

Please refer to the SJAA Website home page (www.sjaa.net) and click on any "Upcoming Event" highlighted in Blue. It will take you to the SJ-Astronomy Meetup page (www.meetup.com/SJ-Astronomy/events/) where you can find specific event times, locations, maps and possible cancellation due to weather.

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Christmas Tree Nebula

Photo by SJAA Member: **Nikola Nikolov**

November 28, 2022

Location: Chile

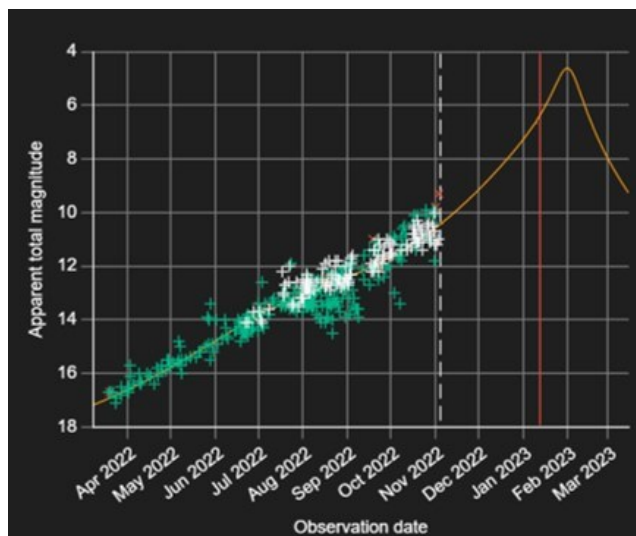


Comet C/2022 E3 (ZTF)

Nikola Nikolov

The first bright comet in 2023 that can be visible with naked eye, C/2022 E3 (ZTF)

Many of us remember the spectacular comet C/2020 F3 “Neowise” during the summer of 2020. Today another bright comet is “promising” similar sky show in the early 2023, the comet C/2022 E3 (ZTF). According to the “Comet Observation Database” (<https://www.cobs.si/>), the comet will reach a maximum magnitude of 4.6 in the beginning of February 2023 and will make it visible to the naked eye.



Comet Observation Database forecast for C/2022 E3 (ZTF).

Credentials: <https://www.cobs.si/>



Comet C2022 E3 (ZTF).

Credit: Nikola Nikolov

What makes C/2022 E3 (ZTF) unique is that when it reaches naked eye brightness the comet will be near Polaris in the winter sky in January 2023. Thus observers and astrophotographers will enjoy the comet most of the time during the long January nights. In December 2022 the comet is accessible for telescope observation and photography after 3 AM in the constellation “Corona Borealis”.

SJAA wishes its members Happy Holidays and Best Wishes for a Happy New Year 2023!



Did You Know?

SJAA is an Annual Supporter of the International Dark-Sky Association!

Credit: Marianne Damon

IDA's purpose is to protect the night sky from light pollution. They work to find solutions that benefit everyone. The IDA is "the recognized authority on light pollution and is the leading organization combating light pollution worldwide." Who besides the astronomy community would be interested in protecting our night sky?

- Birders, concerned about the detrimental effects of artificial lights at night on bird migrations, deaths due to building impacts, and disruption of their biologic clocks.
- Wildlife biologists, concerned about its effects on animal and insect health and behavior. (Example: poor sea turtle hatchlings head towards brightly-lit buildings rather than towards the sea!)
- Physiologists and physicians, concerned about the effects of artificial light on human health, especially for people working night shifts.
- Conservationists, concerned about energy waste and unnecessary emissions from lighting up the night.
- City Planners, concerned about unnecessary costs of unshielded and unnecessary lighting.
- Everyone concerned about safety, because "Brighter does not mean safer".
- And, finally, anyone who has been annoyed by their neighbor's bright lights that stay on all night!

The IDA website www.darksky.org has tips on lighting, public outreach materials, IDA publications, videos, materials for educators, a listing of dark sky places, and a short film, "Losing the Dark". There is a link to the Artificial Light at Night Research Literature Database (no login required), the only one of its kind. Yes, there is even a full page devoted to ways to encourage that neighbor to try alternatives!

In 2017, Andy Kreyche from the Santa Cruz Chapter of IDA came to Houge Park to give us a talk about the organization. A few years ago, we were thrilled to be an audience for the pre-release showing of the film, *SAVING THE DARK* (savingthedark.com), presented in person by Sriram Murali. This hour length film can now also be viewed on YouTube, and our SJAA logo shows up at the very beginning as a co-sponsor! It is a comprehensive documentary that can be watched in segments, and features local sites and persons, such as Chabot Space and Science Center, and scenes filmed at the Golden State Star Party. Sriram Murali also composed *Lost in Light-- a Short Film on Light Pollution*, shot mostly in California and also available on YouTube (<https://youtu.be/HHLnnXgDUgc>). It shows how the Milky Way looks under the different Bortle classes. In a short but powerful 2 minutes and 53 seconds he takes us from an 8 in San Jose, 7 in Mountain View, 6 in Los Altos Hills, 5 in Monte Bello OSP, 4 at Pidgeon Point Lighthouse, 3 at Crater Lake, 2 at both Lassen National Park and Mount Shasta, to the crowning glory, a Bortle 1 sky at Eureka Dunes, Death Valley National Park. Because most urban/suburban people have not seen the Milky Way, as it becomes visible at a 3 or 4 and most suburban skies are 5-6-7 (New York City is a 9!), he decided to make a companion film, *Lost in Light II*, using the more familiar Orion (<https://youtu.be/TVAB1IT-sWw>). Another film, newly being shown as of October of this year, is *Defending the Dark: Preserving the Night Sky in Maine*, by Tara Roberts Zabriskie. The trailer is available on YouTube (<https://youtu.be/OyKuLsuEeTE>).

Because we cherish our night sky, SJAA has elected to be a supporter of IDA. Our Board of Directors has voted each year to contribute \$100, and just recently we have increased our donation to \$500 to support their important work. We have partnered with the local chapter and with Pinnacles National Park for our Pinnacles Star Parties. PNP is currently working towards designation as an International Dark Sky Park, part of a program (International Dark Sky Places) administered by IDA.

Some of us are also individual members. As Andy said, "IDA is a small organization trying to do big things and smart scopes and light pollution filters notwithstanding, it makes sense that all astronomers support the organization and its efforts." Membership benefits include Nightwatch (a monthly e-mail update), a quarterly magazine, and Nightscape, the annual report. We can all do something, both individually and collectively, to decrease night-time light pollution. And sharing the night sky with the public--which we at SJAA do so well!--goes a long way towards building awareness of the importance of curbing light pollution.



Congrats Karen Bieber!

SJAA inspires many, old and young to pursue astronomy and astrophotography as a new hobby, and we've recently had one of our own members Karen Bieber win the Panstarrs Comet Image Contest. Capturing an impeccably well timed shot of K2 PanSTARRS flying by the M10 Globular Cluster, this image surely deserves all the recognition it can get! The image was taken at RCDO on July 14th 2022 with just 7 / 120 sec exposures when K2 PanSTARRS came as close as 1.8 AU (270 million km) from Earth.

"To be honest and to my embarrassment, I had forgotten about entering the contest until I was contacted by Losmandy last month. Although I've been using Photoshop for many years, and image processing is not new to me, I've only been taking astrophotos since April of '21! It was a great honor to be sure, because I still feel like a beginner!"
- Karen Bieber

Contest results were first published in the Losmandy Press Newsletter - Issue 21.
The winning image is below, with technical details on the equipment used for getting the shot.
Karen, heartiest congratulations from everyone at SJAA!



Imaging Telescopes Or Lenses - William Optics Redcat 71

Imaging Cameras - ZWO ASI2600MC Pro

Mounts - iOptron GEM45

Accessories - ZWO ASIAIR Plus

Software - Adobe Photoshop · Pleiades Astrophoto PixInsight

Guiding Telescopes Or Lenses - William Optics UniGuide 50

Guiding Cameras - ZWO ASI290MM Mini



Pinnacles Star Party

Credit: Marianne Damon

SJAA has had a long-term relationship with the Pinnacles National Park (PNP). Being a relatively short drive from San Jose (about 2 hours) and having relatively dark skies, it is a good place to observe. Our leads arrange with the rangers for observing at the West (Soledad) Entrance, where there is no camping. In addition, SJAA members have camped and observed at the East (Hollister) Entrance. At that site we have also arranged and hosted star parties for the public, which are very popular events. These public star parties were suspended during the pandemic.

Vini Carter stepped up to organize the first re-start on April 23rd, to coincide with Dark Sky Week and the Lyrid meteor shower. Ranger Valerie announced the program to the campers. We had seven astronomers who set up in the overflow parking lot. Andy Kreyche from the International Dark-Sky Association gave a talk outside the General Store, and then led people over to the viewing area. After the public left, some stayed up to observe and some went back to their tents. We all were pleased with this effort, but had a "wish list" for the next time.

Ed Wong stepped up to be the lead for future star parties at PNP, continuing his pre-Covid role. The next event was on July 30th. Based on observer feedback to the rangers, a new viewing area had been prepared, away from the glare of headlights. Campers walked down the road and over a bridge to the old Bacon Ranch, where they found a field of telescopes waiting for them. Bob Thomson reported, "It was fun. The transparency was poor but the seeing was excellent. The conditions were the most favorable during the outreach session which was lucky for the public. The conditions got very poor after the public session so no luck there. We got some very good views of Saturn which the public was wowed by and I had the double-double at 333x with a perfectly clean pin-point split with enough black space between the A-B and C-D components that you could drive a truck through it. Excellent seeing."

Our third star party was on August 27. Eleven astronomers participated and there were about a hundred guests. Karen Bieber wrote, "It was an amazingly beautiful night under clear dark skies! Thank you to the organizers who did a great job." Nikhil Tungare reported, "It was a beautiful night indeed and the splendors of the summer milky way put on a dazzling show. Favorites for me from the night were the Swan, Lagoon, Veil, Helix, Dumbbell, Pacman, Double Cluster and the Andromeda Galaxy. All of these showed interesting features that I had not seen in quite some time. The dark lane of Andromeda, the double helix structure of Helix and the filamentary structure of Veil were just mesmerizing. I didn't have a lot of campers stop by my station, but pretty much everyone who did were quite fascinated by the views through the binoculars. Some of the objects that I shared were M22, Lagoon, M7, M24, c/2017 k2 panstarrs, Double Cluster, Andromeda Galaxy, Ink Spot Nebula and ET (Owl) Cluster (Everyone really enjoyed it, once they could see the shape, got a few Awwws...) and the Swan. Thanks to Joe, I had noted my challenge object for the night which was a 12.3+ Mag comet c/2022 E3(ZTF). This was the faintest that I could squeeze out and was able to confirm the observation with the help of Joe's photograph of the comet. In comparison, the 11.6 Mag galaxy NGC6207 besides M13, was easily visible with direct vision. The lit pathway to the observation site was a beautiful touch and I am guessing the most "instagrammed" that evening. Thanks a lot to all the organizers for a memorable event. I am hoping for another one in Sept."

Our fourth star party was on Sept. 24. Ranger Valerie stated that it was the smoothest-running one yet. People other than campers had signed up to come, and there were extra rangers on hand to help out with logistics. Campers and visitors began the evening with a lecture/slide show given by Andy at the campground amphitheater. Marianne set up just above the amphitheater so people who would not be walking over to the telescope field would still be able to have an experience. The entire walk from the amphitheater to the Bacon Ranch was lined with luminaria, and gave a very special feel to the night. The Milky Way was awesome and conditions were ideal. For some of us it was the best sky we have seen this year. There were fourteen astronomers set up to engage with about three hundred viewers. After the star party ended, the astronomers stayed on to do some serious observing and some spectacular astrophotography. Their work is featured in this issue's SJAA Member Astrophoto Gallery.

We were hoping to hold one last star party on October 29, but it was canceled due to the cloudy forecast.

We'll end the season with a quote from our IDA speaker, Andy Kreyche: "I just want to compliment you all on the great experiences you've given to people at Pinnacles over the past few months. It's been a pleasure to meet all (or at least most) of you. In my talk, I give attendees an overview of light pollution, but the star party is the big payoff. What a gift it is that you share your enthusiasm and expertise with them. It's such a great opportunity for visitors to learn new things and have an experience that gives them a cosmic perspective."





Andy Kreyche from IDA getting ready for his talk to campers. The talk is timed to transition from daylight into darkness.



Once campers leave the amphitheater, walk half a mile, and cross over this bridge they experience the wonder of our telescope views.



We are fortunate to have a large area for our vehicles and equipment. Here we are setting up to the right of the Bacon Ranch barn, just as you cross the bridge.



And here we are setting up behind the barn.
See photos taken from this location in this issue's Astrophoto Gallery!





Dates: Sept 24, 2022

Location: East Pinnacles, Paicines, CA

Imaging Telescope: Astro-Tech AT130EDT

Imaging Camera: ZWO ASI294MM-Pro

Mount: iOptron iEQ45 Pro

Software: Pleiades Astrophoto PixInsight, Russell Croman Astrophotography NoiseXTerminator, Russell Croman Astrophotography StarXTerminator, Stefan Berg Nighttime Imaging 'N' Astronomy (N.I.N.A. / NINA)





Dates: September 24, 2022

Location: Pinnacles National Park

Imaging Telescope: WO GT71

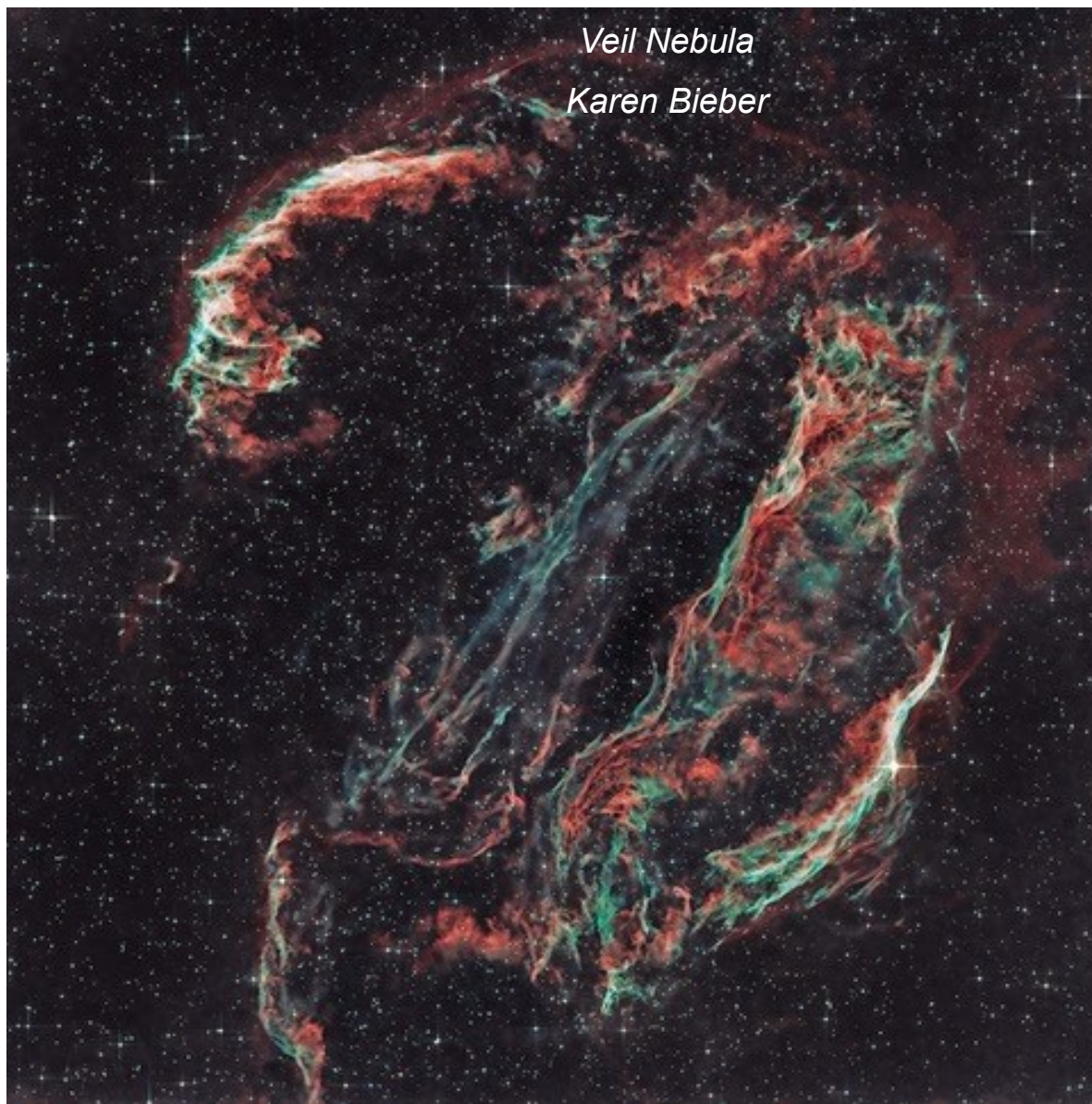
Imaging Camera: ZWO 294MC

Mount: iOptron CEM25p

Software: Pixinsight 1.8, Starnet 2, Gimp



SJAA Member Astrophoto Gallery



Veil Nebula
Karen Bieber

Date: September 24, 2022

Location: Pinnacles National Park

Telescope: RedCat 51

Camera: ZWO ASI2600 MC Pro

Mount: ZWO AM5

Software: None



SJAA Member Astrophoto Gallery



Date: September 24, 2022

Location: Pinnacles National Park

Telescope: Not Applicable

Camera: Canon R6

Mount: Not Applicable

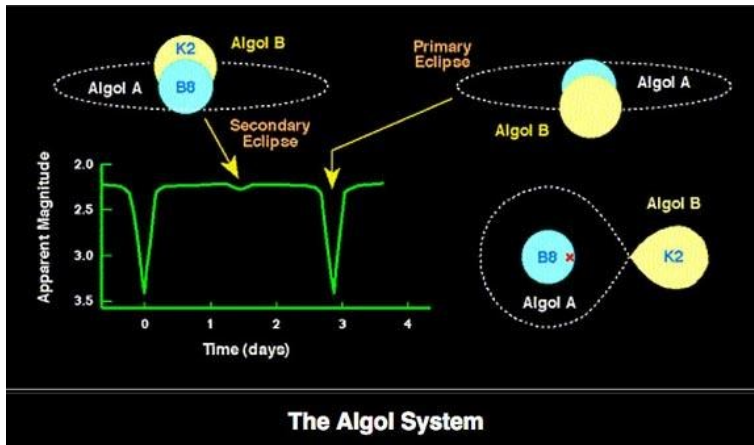
Software: Adobe Camera Raw



Observing Algol an eclipsing binary

Credit: Jordan Makower

Algol, the second brightest star in constellation Perseus (β Persei), is also known as the 'eye of the Medusa', as it appears to brighten and dim over a short period of time. This happens since this star is part of a 3-star system, two of which revolve around each other in our 'line of sight'. Unlike Cepheids, the observed changes here occur when a larger, cooler, and dimmer star blocks the light of the hotter, more luminous 'primary' star. A second eclipse happens when the less luminous companion hides behind the brighter one, but special photoelectric monitoring would be needed to record this slight dimming. A star's brightness is given a *magnitude number, the brighter the star, the lower the number*. In the diagram below, B8 refers to the brighter star.



Sky & Telescope 8/26/20 (Take a Roller Coaster Ride with Algol...)



Sky & Telescope 7/29/06 (The Minima of Algol)

Observations in Algol's brightness can be noted even without a telescope or binoculars. Stars ϵ Persei (magnitude 2.9) and γ Andromedae (magnitude 2.1), alongside β Persei can be used to compare its brightness.

If the observer keeps a chart of Algol's apparent brightness every hour, the reduction in light will be seen; it happens quickly! The top chart indicates how often this occurs. Look on the internet under "Minima of Algol" for local dates and times of these occurrences.



Hubble Views a Billowing Cosmic Cloud

Credit: NASA



Image credit: ESA/Hubble, NASA & STScI, C. Britt, T. Huard, A. Pagan

A small, dense cloud of gas and dust called CB 130-3 blots out the center of this image from the NASA/ESA Hubble Space Telescope. CB 130-3 is an object known as a dense core, a compact agglomeration of gas and dust. This particular dense core is in the constellation Serpens and seems to billow across a field of background stars.

Dense cores like CB 130-3 are the birthplaces of stars and are of particular interest to astronomers. During the collapse of these cores enough mass can accumulate in one place to reach the temperatures and densities required to ignite hydrogen fusion, marking the birth of a new star. While it may not be obvious from this image, a compact object teetering on the brink of becoming a star is embedded deep within CB 130-3.

Astronomers used Hubble's Wide Field Camera 3 to better understand the environment surrounding this fledgling star. As this image shows, the density of CB 130-3 isn't constant; the outer edges of the cloud consist of only tenuous wisps, whereas at its core CB 130-3 blots out background light entirely. The gas and dust making up CB 130-3 affect not only the brightness but also the apparent color of background stars, with stars toward the cloud's center appearing redder than their counterparts at the outskirts of this image. Astronomers used Hubble to measure this reddening effect and chart out the density of CB 130-3, providing insights into the inner structure of this stellar nursery.





Kid Spot Jokes:

Q: Why does the moon orbit the Earth?

A: To get to the other side??

Q: How do astronomers organize a party?

A: They planet.

Q: What music do Astronauts listen to?

A: Nep-Tunes!

Scavenger Hunt: The Moon's Surface

Credit: Jordan Makower



Using binoculars, a small telescope, or a camera with a 300mm telephoto lens, see if you can locate the following surface features, not all of which are shown in this image. Imagine the image to be like a clockface; there is an interesting point at '1 o'clock' in the picture. Refer to the times and dates of your observations.

Your sketches/photos are useful. Each feature listed is described by its appearance.

Lunar Highlands. Light gray, rough appearance. Closer views show many craters, mountains.

Large Craters. Large black circles or smaller bright white circles

Maria/Mare. Dark grey-black circles, uniform color, some overlap others.

Rays/Ray. Straight white lines radiating out of a circle.

Ejecta (blanket). Rough-looking grey area around a crater, may not be symmetrical (around impact crater).

Chain craters. Line of craters, becoming smaller in one direction.

Rilles. May look like a line, like a valley, could be arc-like, or 'S' shaped.

Dome (s). Looks like a bulge, wide, round. Up to 9 miles across. (Moon diameter is 2100 miles)

Newer crater in older crater. Newer one looks brighter.

Mascon (area). Large roundish-irregular black areas may be inside large crater.

History: Moon formed about 4.5 billion years ago, was molten, closer to Earth. Cooled, developed crust, huge, massive impacts caused craters to be flooded with lava.

Some volcanic activity formed domes/craters, atmosphere lost, meteor impacts produced many craters, radiating lines, and ejected material around crater. Fracturing/subsidence formed valleys (rilles). Larger images show greater details and other features.



From the Board of Directors

Announcements

◇ San Jose Astronomical Association is closely monitoring the corona virus COVID-19 situation in Santa Clara County. Since the mask mandate restriction was lifted as of June 3, SJAA has gradually reconvened the in-person activities, such as In Town Star Party, Starry Nights Star Party, and Solar Observation.

◇ Please check our Meetup to determine if an event has been moved online.

◇ New board member elections will be held at the February 2023 board meeting on February 4. The following 4 board positions will be open for Elections: Director 1, Director 3, Director 5, and Director 7. All current incumbents are expected to run for their positions again. Any member interested in running for these positions should reach out to the Secretary at secretary@sjaa.net for eligibility criteria and other details.

◇ New officer elections for President, Vice-President, Treasurer and Secretary positions will be held at the March 2023 board meeting on March 4. Any member interested in running for these positions should reach out to the Secretary at secretary@sjaa.net for eligibility criteria and other details.

Board Meeting Excerpts

September 10, 2022

In attendance

Ken, Rob J, Glenn, Wolf, Vini, Peter, Sukhada, Steven,
Absences: Swami, Aroon
Guests: Dave, Joe, Nancy, Kanch, Leslie

SJAA Volunteer Roles List

To reinvigorate and grow SJAA, the SJAA board discussed a volunteer roles list to enable better recruiting as well as better program leader transitions.

2023 SJAA Calendar

Kanch and Joe did a preview of the 2023 SJAA calendar, and a walk through of the calendar update process with leaders.

October 15, 2022

In attendance

Rob J, Swami, Sukhada, Wolf, Vini, Glenn, Ken, Peter, Aroon
Absences: Glenn, Steven
Guest: Leslie, Kanch

Minors Policy

The board discussed the minors policy, and a code of conduct document is to be sent out to the board for review.

November 12, 2022

In attendance

Ken, Swami, Wolf, Vini, Peter, Glenn, Aroon
Absences: Sukhada, Rob J
Guests: Kanch, Toni-Gregorio Bunch, Joe

Lifetime membership

The board discussed coming up with formal guidelines about what qualifications would constitute towards SJAA offering lifetime memberships to long-time members.

SJAA 2023 Calendar Follow up

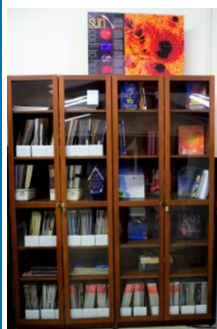
Kanch to discuss the calendar with the board and finalize it so that it can go on the Ephemeris calendar.

Call for Volunteers

You may have noticed in another section of this edition of the Ephemeris that a couple of positions opened up. It would really be good to have those positions filled, so if you might be thinking of getting more involved, please consider it! From the management of the organization (as a member of the board) to a program leader (such as QuickStart), your efforts would help further the mission of the SJAA, which is to bring astronomy and related sciences to the public. There are plenty of ways to get involved, including some that haven't been considered before! If you have an idea that might fit well within the SJAA, please let us know! The best way to reach us is by sending an email to asksjaa@sjaa.net.



SJAA Library



SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers! You may still send all your questions/comments

to librarian@sjaa.net.

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. Headed up by Vini Carter, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

<http://www.sjaa.net/programs/fixit/>

Solar Observing

Solar observing sessions, headed up by Wolf Witt, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change.

<http://www.meetup.com/SJ-Astronomy/>

Quick STARt Program

The Quick STARt Program, headed up , helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick Start sessions are generally held quarterly. The Quick Start program is currently on hold pending a leader stepping up to own the program.

<http://www.sjaa.net/programs/quick-start/>

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, run by David Grover, each with a similar format, with only the content chang-

ing to reflect what's currently in the night sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with views through SJAA Telescopes at the In-Town Star Party, to get a better look at the night's celestial objects.

<http://www.sjaa.net/programs/beginners-astronomy/>

Loaner Program

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met. If you are an SJAA member and an experienced observer or have been through the SJAA Quick STARt program please fill this form to request a particular item. Please also consider donating unused equipment.

<http://www.sjaa.net/programs/loaner-telescope-program/>

Astro Imaging Special Interest Group (SIG)

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. Run by Hy Murveit, the Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

<http://www.sjaa.net/programs/imaging-sig/>

Astro Imaging Workshops and Field Clinics

Not to be confused with the SIG group this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astrophotography, to have a chance of seeing what it is all about.

Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's (Program Coordinator) for additional information.

<http://www.sjaa.net/programs/school-star-party/>

SJAA Contacts

President/Dir:	Ken Miura
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Treasurer/Dir:	Rob Jaworski
Secretary:	Aroon Nair
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Director:	Vini Carter
Director:	Swami Nigam
Director:	Glenn Newell
Director:	Wolf Witt
Director:	Sukhada Palav
Director:	Peter Melhus
Ephemeris Newsletter -	
Editor:	Abishek
Prod. Editor:	Emi Nikolov
Binocular Stargazing:	Ed Wong
Fix-it Program:	Vini Carter
Hands on Imaging:	Glenn Newell
Imaging SIG:	Hy Murveit
Intro to the Night Sky:	David Grover
In-Town Star Party:	Muditha Kanchana
Library:	Jai Purandare
Loaner Program:	Muditha Kanchana
Memberships:	Kaushal
Publicity:	Beth Johnson
Quick STARt	Vacant
Donations:	Rich Gregor
Solar:	Wolf Witt
Starry Nights:	Joe Fragola
School Events:	Jon Anderson
Refreshments:	Marianne Damon
Social Manager:	Beth Johnson
Speakers:	Sukhada Palav
Webmaster:	Satish Vellanki

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles, including Observation Reports for publication should be submitted by no later than the 20th of the month of February, May, August and November.

San Jose Astronomical Association
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San Jose, CA 95159-8243
<http://www.sjaa.net/contact>

Place
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here

San Jose Astronomical Association
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San Jose, CA 95159-8243

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San Jose Astronomical Association Annual Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

Membership Type (must be 18 years or older):

☐ **New** ☐ **Renewal**

☐ **\$20** Regular Membership with online Ephemeris

☐ **\$30** Regular Membership with hardcopy Ephemeris mailed to below address

The newsletter is always available online at: <http://www.sjaa.net/sjaa-newsletter-ephemeris/>

Questions? Send e-mail to: memberships@sjaa.net

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at: <http://www.sjaa.net/join-the-sjaa/>

Name:

Address:

City/ST/Zip:

Phone:

E-mail address:

